

# ALAN LAIRD

Senior SRE & Infrastructure Engineer | Terraform · Kubernetes · Ansible · GPU & HPC · Observability

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## PROFESSIONAL SUMMARY

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20+ years building and operating large-scale infrastructure across GPU compute platforms, HPC environments, and enterprise SaaS. Expert in Terraform and Kubernetes end-to-end — bare metal provisioning through production operations — and deep in Linux systems internals, networking (L3/L4, BGP, DNS, VPC), and distributed systems. Proven in building observability stacks (Prometheus, Grafana, Loki, Tempo), CI/CD-driven deployment pipelines, and cloud automation across AWS and GCP. Hands-on experience with GPU fleet automation and NCCL benchmarking on H100/L40S clusters; HPC scheduler background (TORQUE, PBS, LSF, SGE; Slurm-adjacent). Bash scripting for tooling and automation.

## CORE SKILLS

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**IaC & Automation:** Terraform, Ansible, Puppet, Packer, Salt

**Cloud & Virtualization:** AWS (VPC, EC2, CloudFormation, Security Groups), KVM/QEMU/Libvirt, OpenStack, VMware

**Containers & Orchestration:** Kubernetes (bare metal & cloud, multi-node, GPU workloads), Docker, CoreOS, Talos Linux, Helm, Kustomize, ArgoCD (GitOps)

**CI/CD & Pipelines:** Jenkins, ArgoCD/GitOps, GitHub Actions, OS release pipelines, Docker image cadences

**Observability:** Prometheus, Grafana, Loki, Tempo, Phlare, VictoriaMetrics, AlertManager, MetalLB, distributed tracing, continuous profiling

**OS & Networking:** Linux (Ubuntu, CentOS, CoreOS, RHEL), Classic & SDN Networking, BGP, Consul, Syslog

**Security & Compliance:** ISO 27001 (Certified Trainer), PCI Level 2, systems hardening, audit support

**Other:** HPC schedulers (TORQUE, PBS, LSF, SGE — Slurm-adjacent), Bash, Python (reading/review), SAN/distributed storage (Cassandra, VAST/NFS), F5 load balancing

## WORK EXPERIENCE

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### Lead Senior Infrastructure Engineer — Compute Platform | Akamai

June 2023 – Present Remote (Boston, MA)

Led infrastructure automation for an internal IaaS platform serving GPU and general compute workloads, accelerating feature delivery and enabling rapid GPU fleet expansion across production environments.

- Bootstrapped a cloud environment on KVM/Libvirt and Ansible to deliver the team's first internal cloud platform, reducing development risk and increasing feature velocity.
- Integrated platform services against a large existing Salt configuration management base.
- Drove service-oriented engineering culture and best practices across a 20-person infrastructure team.

### Infrastructure Developer — Engineering Shared Services | Viasat

October 2021 – March 2023 Remote (Carlsbad, CA)

SRE and DevOps role maintaining customer-facing and global engineering core services and pipelines for a leading consumer and military satellite internet provider.

- Maintained and extended complex Ansible playbooks for system provisioning and service automation across global infrastructure.
- Rebuilt and modernized Ubuntu and CentOS real-time kernel build pipelines using EI-repo and RT sources.
- Designed and deployed an Ansible-managed jump box service in AWS using Terraform for VPC and Security Groups, and Jenkins for automated weekly rebuilds — significantly improving engineering platform security.
- Modernized monthly Docker releases for OS and platform pipelines; supported Python components of the Jenkins CI/CD environment.
- On-call support and troubleshooting for complex Consul and Syslog routing environments.

## **Systems Manager / SRE / Enterprise Systems Architect | Iovation (Anti-fraud SaaS — clients include Starbucks, FedEx)**

*February 2012 – October 2021 Remote (Portland, OR)*

Owned all infrastructure for a high-availability anti-fraud SaaS platform processing transactions for Fortune 500 clients. Built and led the systems team through nearly a decade of growth, datacenter modernization, and global expansion — maintaining high-nines availability throughout.

- Rebuilt multiple datacenters from legacy Sun/Hitachi hardware to Dell/Supermicro CentOS environments fully managed with Puppet IaC; scaled to 64 KVM compute hosts supporting ~1,000 virtual machines — lowering cost and dramatically increasing delivery speed.
- Abstracted Cassandra and other customer-facing systems; built new physical datacenter locations in Miami and Amsterdam; added hybrid cloud patterns to support Asia-Pacific expansion.
- Built Kubernetes greenfield bare metal environments to validate different High Availability and top-of-rack BGP patterns.
- Engineered 50-node OpenStack and Kubernetes environments with OS release pipelines using Packer.
- Built and automated MVP Docker versions of the product stack using Ansible on AWS, CoreOS, and Kubernetes.
- Improved SRE tooling for automation, monitoring, alerting, and IaC in Ansible, Terraform, and AWS CloudFormation.
- Established monthly OS release cadence producing security-patched base Docker images for all platform services.

## **Sr. SaaS Operations Engineer | Jive Software**

*December 2010 – February 2012 Portland, OR*

- Navigated complex infrastructure work supporting high-value customers on the Jive SaaS platform.
- Wrote Puppet code managing large numbers of AWS and VMware virtual systems, networks, and security groups.
- Drove process creation to increase accuracy and throughput while onboarding staff and doubling team size.

## **HPC UNIX Systems Administrator | Penguin Computing**

*January 2010 – December 2010 Remote (Fremont, CA)*

- Provided UNIX admin support for the Bioscope genetic sequencing product; managed bioinformatics customers on HPC Linux and Solaris clusters (TORQUE, PBS, MAUI, SGE, LSF).
- Modernized and significantly increased HPC cluster performance for NIH, Sorbonne, and other research institutions.

## **Network and Systems Architect | Musician's Friend**

*June 2002 – December 2009 Medford, OR*

- Designed and built hundreds of UNIX systems, SAN storage, datacenters, and LAN/WAN networks including load balancers, firewalls, and caching layers; hired and trained engineering staff.
- Architected network and systems security controls leading to PCI Level 2 compliance; led ISO 27001 certification effort.
- Architected infrastructure capacity and efficiency improvements that contributed directly to 8x revenue growth over the period.
- Led Akamai global load balancing integration with layer-7 traffic shaping and active website frame management for tiered load response.
- Awarded Innovator of the Year (2006) in recognition of infrastructure and platform contributions to the company's growth.

## **RECENT PROJECTS & TECHNICAL WRITING**

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### **GPU Cluster Benchmarking on Nebius — Terraform + Ansible + NCCL**

[github.com/alanlaird/gpulabs](https://github.com/alanlaird/gpulabs) | [alan.laird.net/tech/2026/03/04/nebius-gpu-benchmarking.html](https://alan.laird.net/tech/2026/03/04/nebius-gpu-benchmarking.html)

- Built a toolkit to provision GPU nodes on Nebius AI Cloud (H100/L40S), run NCCL all-reduce benchmarks, and tear everything down from a single make command.

- Terraform handles instance provisioning with auto-selected CUDA images; Ansible installs Docker, NVIDIA Container Toolkit, and configures inter-node MPI/SSH. Supports both TCP and 400 Gbps InfiniBand fabrics with runtime auto-detection.
- Includes force-destroy handling for stuck Terraform state on IB nodes and H200 support in eu-west1.

### **Talos Linux Kubernetes on GCP Spot Instances with MIG Auto-Healing**

[github.com/alanlaird/gcplabs](https://github.com/alanlaird/gcplabs) | [alan.laird.net/tech/2026/03/07/spot-talos.html](https://alan.laird.net/tech/2026/03/07/spot-talos.html)

- Terraform + Makefile project provisioning an immutable Talos Linux Kubernetes cluster on GCP spot instances, with Managed Instance Groups providing automatic node recovery on preemption.
- Uses a reserved static IP so the K8s API endpoint survives control plane replacement; machine configs embedded as user-data in instance templates so replacement nodes self-configure and rejoin without manual intervention.
- Includes live cost estimation (~\$24/month for 3-node cluster), preemption history, and full cluster restore in ~10 minutes.

### **incus-lab — Local Virtualization Lab with Terraform**

[github.com/alanlaird/incus-lab](https://github.com/alanlaird/incus-lab)

- Terraform-managed local lab environment using Incus (LXD successor) for developing and testing infrastructure automation workflows locally before cloud deployment.

### **Kubernetes Observability Playground — Prometheus · Loki · Tempo · Phlare · VictoriaMetrics · ArgoCD**

[github.com/alanlaird/kind-monitoring-prometheus-loki-tempo-phlare-victoriametrics-grafana-playground](https://github.com/alanlaird/kind-monitoring-prometheus-loki-tempo-phlare-victoriametrics-grafana-playground)

- Full-stack observability environment running on KinD with ArgoCD GitOps delivery — covers metrics (Prometheus, VictoriaMetrics), logs (Loki), distributed tracing (Tempo), continuous profiling (Phlare), and custom Grafana dashboards for cluster capacity, ArgoCD health, cert-manager, MetalLB, and more.
- Includes MetalLB with automated iptables DNAT script for LAN-accessible LoadBalancer services from bare metal KinD clusters; dummy metrics service for dashboard validation.
- Grafana dashboards provisioned via Kustomize ConfigMaps; ArgoCD sync-wave ordering ensures correct dependency sequencing across 10+ namespaces.

## **EARLY CAREER**

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IT Manager / Lead Integrator — Maverick Networks / Broadcom (San Jose, CA, 1998–2001): Primary engineering support for startup and through Broadcom acquisition; responsible for systems, networks, labs, facilities, and ISO 9001 efforts. Integrated software dev environments for multiple Silicon Valley acquisitions.

Engineering System Admin / IT Manager — Silicon Graphics (Mountain View, CA, 1997–1998): Supported four labs, hundreds of SGI systems, complex ATM networks, and three software engineering teams. Developed test harnesses driving performance application tuning guides that helped eclipse all other UNIX vendors in Usenet NEWS server sales in 1998.

Earlier roles: IT Manager at Taos Mountain Software (Palo Alto); ISP founder and systems/network builder at InfoStructure (Ashland, OR); tools developer and game designer at Trilobyte — The 7th Guest (Jacksonville, OR).

## **EDUCATION & CERTIFICATIONS**

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**Computer Science coursework** — Southern Oregon University

**ISO 27001 / BSI 1799 Certified Trainer** — 2006